

Fig. 6: Pin details of 7805, 7812 and BC547

exit gates, respectively.

LM358 op-amp. Dual-operational amplifier LM358 (IC4) is used as a voltage comparator to compare the output of the IR sensors with a fixed threshold voltage in order to know whether the IR beam is interrupted or not.

IR transmitter and receiver. Two IR transmitter-receiver pairs are used. The IR LEDs are connected in forward-biased condition to the +5V power supply through 220-ohm resistors. These emit IR light, which is interrupted when an object comes into its way to the IR receiver. The IR

receiving photodiodes are connected in reverse-biased condition to +5V power supply through 1-mega-ohm resistors. When the IR light falls on the photodiodes, their resistance changes and so does their output. This output is compared with a fixed voltage to give a digital output to the microcontroller in order to judge the entry and exit of the vehicles.

LCD display. LCD1 is a two-line, 16-character, alpha-numeric liquid crystal display. Data lines D0 through D7 of the LCD are connected to port 2 of AT89S52 (IC3). Reset (RS) and enable (E) control lines are connected to port pins P3.6 and P3.7, respectively. Control lines control data flow from the microcontroller to LCD1.

When power is switched on, LED1 glows to indicate the presence

of power in the circuit and LED2 glows to indicate the presence of RFID reader. Simultaneously, the 'Automatic RFID Car Parking' message is displayed on LCD1 along with a short beep from piezobuzzer PZ1. Transistor BC547 drives the buzzer. Pin details of 7805, 7812 and BC547 are shown in Fig. 6.

When a car crosses the IR LED1-D1 pair installed at the entry gate, the gate boomer does not open until an RFID tag is placed near the RFID reader. After the tag is placed near the reader, the gate boomer opens for three seconds and closes automatically. If the initial recharge amount was Rs 900, the LCD display shows 'Vehicle! Amount' in the first line and 'Deducted 100' in the second line, followed by 'Balance Amount' in the first line and '800' in

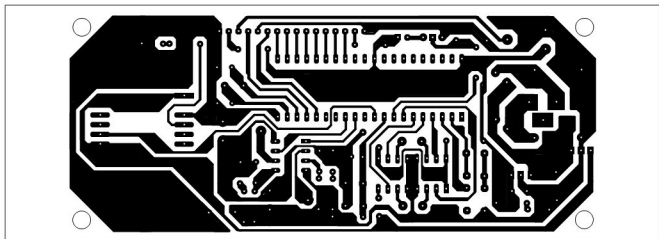


Fig. 7: An actual-size, single-side PCB for the RFID-based automatic vehicle parking system

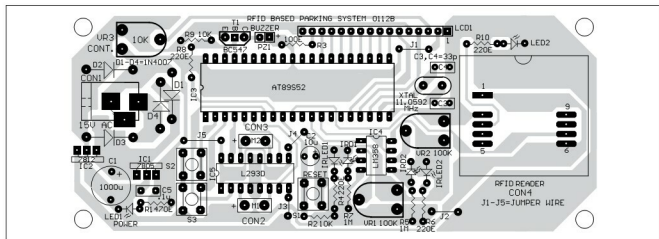


Fig. 8: Component layout for the PCB